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ELE 443 - Control Systems Lab

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# Lecture 2: Mathematical Operations, Random Numbers, and Symbolic Math

# Addition & Subtraction

- Addition and subtraction are performed between matrices having the same dimensions.

  - A=[7 5 9;2 3 6]

```
A =
     7     5     9
     2     3     6
```

  - B=[4 8 1;3 5 4]

```
B =
     4     8     1
     3     5     4
```

  - C=A+B

```
C =
    11    13    10
     5     8    10
```

Must have the same dimensions

D=A-B

```
D =
     3    -3     8
    -1    -2     2
```

# Array Multiplication

- Matrix multiplication is defined for matrices A and B such that the number of columns of A is equal to the number of rows of B.
- Let A be m-by-n matrix and B is a p-by-q matrix. The matrix multiplication A*B is defined iff n=p.
- In this case, the resulting **matrix C=A*B is an m-by-q matrix.**
- **Note that matrix multiplication** is not commutative.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \qquad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$

$$A * B = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \end{bmatrix}$$

# Array Multiplication

- Consider the following Example:

  - A=[7 8 9;3 2 8;5 4 -2];
    B=[1;-9;3];

    C=A*B
    C = -38
        9
        -37

A(3x3)*B(3x1)
=C(3x1)

# Inverse of a matrix

- The inverse of matrix M is denoted by $M^{-1}$ and is defined such that $M*M^{-1}= M^{-1}*M=I$ where I is the identity matrix.
- M is invertible (i.e. $M^{-1}$ exists) iff M is nonsingular (i.e. it has a nonzero determinant, $\det(M)\neq 0$)
- The inverse of M is calculated using the command inv(M).
- Consider the following examples

```
A=[1 0 0;0 2 1;2 -2 1]
A=  1    0    0
    0    2    1
    2   -2    1
det(A)
ans =4
```

```
inv(A)
Ans=   1       0       0
     0.5     0.25   -0.25
      -1      0.5     0.5
A*inv(A)
ans =  1    0    0
       0    1    0
       0    0    1
```

# E-B-E Operations

- Element by element (E-B-E) operations are performed by typing a dot (.) before the operator.
- Addition and subtraction are E-B-E operations.
- E-B-E multiplication, division and power are done between matrices having the same dimensions.

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \quad B = \begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix}$$

$$A.*B = \begin{bmatrix} a_1 b_1 & a_2 b_2 & a_3 b_3 \end{bmatrix}$$

$$A./B = \begin{bmatrix} a_1 / b_1 & a_2 / b_2 & a_3 / b_3 \end{bmatrix}$$

$$A.\hat{}\,B = \begin{bmatrix} a_1^{b_1} & a_2^{b_2} & a_3^{b_3} \end{bmatrix}$$

# Array Analysis

| Function | Description | Example |
|---|---|---|
| mean(V) | For vectors, MEAN(V) is the mean value of the elements in V. If V is a matrix, then MEAN(V) will return a row vector containing the mean of each column. | V=[5 8 9 10];<br>mean(V)=8<br>A=[1 3;2 1]<br>mean(A)=[1.5 2] |
| mx=max(M)<br>mn=min(M) | For vectors, mx (mn) is the largest (smallest) element in X. For matrices, mx (mn) is a row vector containing the maximum (minimum) element from each column. | M = 2 3 4<br>5 8 9<br>-1 0 7<br>max(M)<br>ans =5 8 9 |
| rank(M) | Provides an estimate of the number of linearly independent rows or columns of a matrix A. | M=[1 7 8;2 6 4];<br>rank(M)<br>ans =2 |

# Array Analysis

| Function | Description | Example |
|---|---|---|
| [V,D] = eig(X) | produces a full matrix V whose columns are the corresponding eigenvectors and a diagonal matrix D of eigenvalues such that A*V=V*D.<br>The eigenvectors in V are normalized so that each has a norm of one. | A=[1 3;2 1]<br>[V,D]=eig(A)<br>V = 0.7746 -0.7746<br>0.63246 0.63246<br>D = 3.4495 0<br>0 -1.4495 |
| sum(X) | is the sum of the elements of vector X. | X=[1 2 3];<br>S=sum(X)<br>S =6 |

# Array Analysis

| Function | Description | Example |
|---|---|---|
| expm(M) | is the matrix exponential of the square matrix X. | M=[2 1;5 -4];<br>expm(M)<br>ans =13.9761 2.0718<br>10.359 1.5453 |
| exp(M) | computes the exponential of X element-by-element | M=[2 1;5 -4];<br>exp(M)<br>ans = 7.3891 2.7183<br>148.41 0.018316 |

# Array Analysis

- Note that matrix exponential (expm in MATLAB) is computed according to Taylor series expansion of exponential function, where the operand is a matrix and not a scalar.

$$\exp(M) = I + M + \frac{M^2}{2!} + \frac{M^3}{3!} + \ldots = \sum_{i=0}^{\infty} \frac{M^i}{i!}$$

- That's why expm is only applied to square matrices because it's a linear combination of power of matrices.

# Array Analysis

- **std(V)**: returns the standard deviation of the elements of vector V.
- **median(V)**: For vectors, median(V) is the median value of the elements in V.
- **sort(X)**: For vectors, it sorts the elements of X in ascending order. For matrices, it sorts each column of X in ascending order.
- **det(M)**: returns the determinant of the matrix M.
- **dot(V1,V2)**: returns the scalar product of the vectors V1 and V2
- **cross(V1,V2)**: returns the cross product of the vectors V1 and V2.

# Random numbers

- "**rand**" function generates uniformly distributed pseudo-random numbers.
  - random numbers are between 0 and 1.
- **rand(N)**
  - returns an N-by-N matrix containing pseudo-random values drawn from a uniform distribution on the unit interval i.e. [0;1].
- **rand(M,N)**
  - returns a random M-by-N matrix

- **randi:** Generates matrix of uniformly distributed random integers.
- **randi([a,b],M,N)** generates an M-by-N matrix in a given range (between a and b, a and b included).

# Symbolic Math

- It defines variables that don't necessarily have a defined scalar or numerical value.
- Symbolic objects can be used as independent variables.
- For example, we can use the command syms such that:
  - syms x y z
  - Therefore x, y and z are three Symbolic Objects.

- A symbolic object can be:
  - A variable with no pre-assigned numerical value.
    - Ex: syms x y t
  - An expression made of symbolic variable/numbers.
    - Ex: syms x y; z=sqrt(x+y)

# Creating Symbolic Objects

- We use the command **`syms`** to create symbolic objects.
- A symbolic expression is a mathematical expression made of one or several symbolic objects.
- To create one Symbolic Object:
  - a=sym(2)
    - a has a symbolic-numerical value
  - b=sym('gamma')
    - b has a symbolic-string value.
- To create multiple symbolic objects:
  - syms x y z

# Symbolic vs Numerical

- Symbolic objects: a and b

```
a=sym(1);
b=sym(2);
f=a/b
f = ½
```

- Numerical objects: A and B:

```
A=1;
B=2;
F=A/B
F =0.500
```

# Symbolic to Numeric

- Some symbolic expressions can have numerical values resulting from numerical operations.
- To convert from symbolic to numerical objects, we use the command **`double(S)`**.
- Example:

```
a=sym(3);
b=1/a
b  = 1/3

B=double(b)
B = 0.3333
```

# subs command

- To substitute a variable in a symbolic expression we use the command **subs**. Consider the following example:
  - syms x y;
    f=2*x+log(y);
    subs(f,x,2);
    ans = 4+log(y)
    subs(f,[x y],[2 1])
    ans = 4

  - syms a b;
    subs(cos(a) + sin(b), [a,b], [sym('alpha'), 2])
    ans =
    sin(2) + cos(alpha)

# Solving equations (Symbolic)

- Solving algebraic equations.
- Example: quadratic equation
  - syms a b c x

    eqn = a*x^2 + b*x + c == 0

    S = solve(eqn)

- Example: System of equations
  - syms x y

    eqns = [2*x + y == 0, x – y == 1];

    S = solve(eqns, [x y])

    x: 1/3

    y: -2/3

# Calculating Derivatives

- Partial Derivatives. Consider the function given by:
  - f = sin(x^2+y)

- Partial derivative w.r.t. x
  - f_x=diff(f,x)
    f_x =2*cos(x^2+y)*x

- 2nd Partial derivative w.r.t. x
  - f2_x=diff(f,x,2)
    f2_x = -4*sin(x^2+y)*x^2+2*cos(x^2+y)

# Calculating Integrals

- Integration. Consider the function given by:
  - f=sin(x)*exp(-x)

- Definite integral (i.e. Integrate f(x) over the range [0,2])
  - F=int(f,x,0,2)
    F =-1/2*exp(-2)*cos(2)-1/2*sin(2)*exp(-2)+1/2

- Indefinite Integral
  - G=int(f,x)
    G =-1/2*exp(-x)*cos(x)-1/2*sin(x)*exp(-x)

# Solving Differential Equations

- Solve a **first-order** differential equation:
- Solve $\dot{x} = ax$
  - syms x(t) a

    eqn = diff(x,t) == a*x;

    S = dsolve(eqn)

    S = C1*exp(a*t)

where C1 is a constant

- Solve a **second-order** differential equation:
- Solve $\ddot{x} = ax$
  - syms x(t) a

    eqn = diff(x,t,2) == a*x;

    S = dsolve(eqn)

    S = C1*exp(-a^(1/2)*t) + C2*exp(a^(1/2)*t)

where C1 and C2 are constants

# Solving Differential Equations

- Solve a second-order differential equation with ICs:
- Solve $\ddot{x} = a^2 x$. ICs: $x(0) = b$ and $\dot{x}(0) = 1$
  - syms x(t) a b

    eqn = diff(x,t,2) == a^2*x;

    Dx = diff(x,t);  % specify the second initial condition

    cond = [x(0) ==b, Dx(0) == 1];

    S = dsolve(eqn,cond)

    S =

    (exp(a*t)*(a*b + 1))/(2*a) + (exp(-a*t)*(a*b - 1))/(2*a)